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Selection, application, and pathogenicity of naturally occurring *Beauveria bassiana* strains against *Ips duplicatus* (Coleoptera: Curculionidae, Scolytinae)

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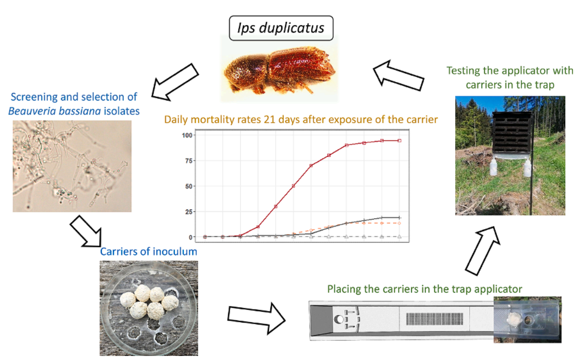
HIGHLIGHTS

- Local *B. bassiana* strains isolated from naturally encountered *I. duplicatus* populations.
- Innovative application of selected *B. bassiana* strains in pheromone traps for pest management.
- *I. duplicatus* infected via traps containing carriers with *B. bassiana.*
- Trap carriers maintain high virulence of *B. bassiana* for up to 21 days.

GRAPHICAL ABSTRACT



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ABSTRACT

Norway spruce (*Picea abies*) stands in Central Europe are increasingly threatened by the northern bark beetle, *Ips duplicatus*, which is a pest whose distribution and impact have expanded significantly in recent decades. The reduced efficacy of traditional pest control methods against this species underscores the need for innovation management strategies. This study investigates a novel approach that combines modified pheromone traps with the biological control agent *Beauveria bassiana*, an entomopathogenic fungus. A total of 48 *B. bassiana* strains were obtained from natural populations of *I. duplicatus*. Following laboratory testing, two strains (NRID11 and NRID43) were selected for field trials. Carriers of the living organism were inoculated with these strains and then placed in applicators as part of the pheromone traps. The study evaluated the effects of selected strains and different carrier exposure periods on beetle mortality rates. The NRID11 strain showed some percentages exceeding 90 % after 21 days of exposure. However, efficacy decreased to less than 60 % when the carriers remained in the traps for more than 30 days. These findings emphasize the importance of optimizing exposure duration and confirm the potential of carriers inoculated with *B. bassiana* in modified pheromone traps as a promising biological control strategy against *I. duplicatus*.

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1. Introduction

The double-spined bark beetle *Ips duplicatus* Sahlberg (Coleoptera: Curculionidae, Scolytinae) is originally from Fennoscandia, Siberia, and East Asia (Wood and Bright, 1992; Pfeffer and Knížek, 1995) and was first reported in Central Europe at the beginning of the 20th century (Holuša and Grodzki, 2008). Currently, *I. duplicatus* is considered an invasive bark beetle, extending southwards from its natural distribution in the north of the continent (Olenici et al., 2010). It occurs in 22 European countries (CABI, 2021) with recent records in Switzerland, Liechtenstein (Wermelinger et al., 2020), and Slovenia (Kavčič et al., 2023). It has a wide range of coniferous hosts, including representatives of the genera *Picea, Pinus, Abies, Larix, Juniperus* (Pfeffer, 1995) as well as *Pseudotsuga menziesii* (Mirb.) Franco (Kašák and Foit, 2015).

Although *I. duplicatus* is a less important pest in its native countries (Mrkva, 1994; Krokene and Solheim, 1996), it has become a serious one of *Picea abies* (L.) H. Karst. in Central Europe (Holuša et al., 2010; Olenici et al., 2010; Grégoire et al., 2024). The double-spined bark beetle mainly attacks 40 to 70-year-old standing Norway spruce trees in managed forests (Grodzki, 2003). The adult beetles fly to the crown part of the stem (Schlyter and Anderbrant, 1993), which makes it difficult to detect early symptoms of an attack. *Ips duplicatus* infests trees that are individually dispersed in the stands, without forming patches (Holuša et al., 2010), and can produce as many as three generations in a single growing season (Holuša et al., 2003; Holuša and Knížek, 2007; Šotola et al., 2021). It prefers trees that are weakened by drought and infected by *Armillaria* spp. (Holuša, 2004; Vakula et al., 2021).

Its economic significance in Central Europe is considerable. Each year, about 10–50 thousand cubic meters of spruce timber infested by *I. duplicatus* are harvested in Slovakia, making it the second most major bark beetle pest after *Ips typographus* (L.) (Vakula et al., 2018). In the Czech Republic, there was a sharp increase in the volume of trees infested by *I. duplicatus* in 2003, and since that time, the volume has exceeded 30,000 m^3 every year (Holuša et al., 2010).

Ips duplicatus was not a significant pest in the past, and forest protection methods against it still lack refinement and are less effective (Lubojacký et al., 2018; CABI, 2021). At present, the most effective protective measures against this species include sanitary logging, the removal of infested trees, and the use of pheromone traps (EFSA, 2017). Sanitary logging is a less efficient approach because the symptoms of *I. duplicatus* attacks are difficult to recognize. Furthermore, most of the *I. duplicatus* beetles, 90.5 % of them, hibernate in forest litter and soil (Anyśko and Starzyk, 2011), while Davídková et al. (2023) state that only 2.2 % of adults overwinter under tree bark, making this method effective primarily during the vegetation period. Unlike other bark beetles, *I. duplicatus* is unable to colonize fallen trees; therefore, the use of trap trees is not effective for this species (Šotola et al., 2021). Other devices used as protective measures include pheromone traps (Duduman et al., 2022). However, as research on *I. typographus* has shown, they capture only a small part of the population; up to 10 % can be caught with high trap density (Weslien and Lindelöw, 1990; Lobinger and Skatulla, 1996).

Entomopathogenic fungi of the genus *Beauveria* (Ascomycota: Hypocreales) are commonly found as insect parasites in terrestrial ecosystems and play an important role in the natural control of arthropod populations. They are generalist entomopathogens with a life cycle characterized by both insect-pathogen interactions and a saprotrophic phase (Khan et al., 2012; Vega et al., 2012). *Beauveria* spp. have been reported to infect a broad range of arthropods both above ground and in the soil environment (Zimmerman, 2007; Uma Devi et al., 2008; Rehner et al., 2011; Lacey et al., 2015). The fungal infection caused by *Beauveria* spp., referred to as white muscardine, appears in bark beetle populations at low, but constant, prevalence levels (e.g. Wegensteiner et al., 2015a; Barta et al., 2018a, 2020; Hyblerová et al., 2021). In Europe, several studies have focused on assessing the prevalence of *Beauveria* during outbreaks of the European spruce bark beetle, *I. typographus* (e.g. Landa et al., 2001; Wegensteiner, 2007; Draganova et al., 2010; Mudrončeková et al., 2013; Wegensteiner et al., 2015b; Barta et al., 2020; Milosavljević et al., 2021), but information on the natural occurrence of these fungi in *I. duplicatus* populations is limited with just a few notes (e.g. Dinu et al., 2012). Four *Beauveria* species, *B. bassiana* (Bals.-Criv.) Vuill., *B. caledonica* Bissett & Widden, *B. pseudobassiana* S.A. Rehner & R.A. Humber, and *B. brongniartii* (Sacc.) Petch, have been detected in populations of bark beetles (Landa et al., 2001; Draganova et al., 2010; Barta et al., 2018a, 2020; Hyblerová et al., 2021), and only *B. bassiana* was identified from *I. duplicatus* in native habitats (Dinu et al., 2012) or infected *I. duplicatus* in controlled laboratory settings (Vakula et al., 2019).

The potential of *Beauveria* spp. as mycoinsecticides against agricultural and forest pests has been extensively researched for many years (Lacey et al., 2015). Considerable progress has been made in developing various mass-production systems of fungal inocula and formulating mycoinsecticides tailored to specific needs in controlling target insect pests (Jaronski and Mascarin, 2017). The use of *Beauveria* species in pest control is considered to be safe (Zimmerman, 2007), and several have undergone testing for their effectiveness in controlling bark beetles (e.g. Vaupel and Zimmermann, 1996; Landa et al., 2001; Kreutz et al., 2004a, 2004b; Draganova et al., 2010, Jakuš and Blaženec, 2011; Mudrončeková et al., 2013; Grodzki and Kosibowicz, 2015; Barta et al., 2018b). Virulence of fungal isolates toward different hosts considerably varies and it is often related to the release of exoenzymes (proteases, lipases, and chitinases) during the infection process (Khan et al., 2012). Selecting highly virulent strains for a target insect host is an important assumption for the successful implementation of the fungus in biocontrol strategies (Barta et al., 2018b). Highly virulent strains are characterized by their capacity to cause the host's death with only a small number of spores within a relatively short time. Although there is currently no microbial insecticide derived from these fungi specifically designed against bark beetles, it is widely believed that they possess the capability to be effectively integrated into a pest management system aimed to control these bark beetle infestations.

The limited range of effective pest control methods makes forest protection against *I. duplicatus* undermined. One potential innovative way involves combining biocontrol and biotechnical approaches, specifically using pheromone traps alongside a biological agent based on *B. bassiana*. This alternative offers a highly selective method by using a bioinsecticide for plant protection, also known as the autoinoculation procedure. Various combinations of devices and entomopathogenic fungi have been tested in the laboratory and in field experiments for controlling different insect pests (Vega et al., 1995; Klein and Lacey, 1999; Maniania, 2002). Grodzki and Kosibowicz (2015) tested the application of *B. bassiana* preparations, commercially produced against *I. typographus* with pheromone traps. The study revealed no significant direct effect of fungi on the beetle population; instead, an indirect one was noted in reducing tree mortality due to bark beetle infestations. The challenges posed by certain factors, such as drought, high temperatures, or rainwater, which negatively affect the survival of *Beauveria* inoculum in traps, remain unresolved (Vakula et al., 2012). Vakula et al. (2019) demonstrated that *B. bassiana* conidia could be transmitted from the parent to the offspring generation of *I. duplicatus*. However, the conidia were only capable of infecting beetles from the offspring generation under controlled laboratory conditions, such as Petri dishes, and not in their natural habitats within bark galleries. Another study demonstrated that the application of *B. bassiana* preparations in the field can result in a 93 % mortality rate. Significant reductions were observed in experiments to infect pheromone traps, including a decrease in the number of bore holes and the length of maternal galleries. Moreover, no larvae, pupae, or juveniles were found under the bark of spruce trunks (Kreutz et al., 2004a).

The objectives of this research were to: (i) sampling and isolation *Beauveria* species from natural populations of *I. duplicatus* in affected Norway spruce trees in the Western Carpathians; (ii) assess their enzymatic activity, pathogenicity, and sensitivity to monoterpenes under laboratory conditions; and (iii) evaluate selected strains for their virulence capacity against *I. duplicatus* adults in field experiments, using a trap applicator equipped with a carrier.

2. Material and methods

2.1. Study area

The study area was located in the northwestern region of Slovakia (49.329337 N, 18.795401 E, 390 m a.s.l.). Orographically, belongs to the Western Carpathians. Geologically, the region is located in the Paleogene flysch. In the region, the Norway spruce (*P. abies*), common beech (*Fagus sylvatica* L.), silver fir (*Abies alba* Mill.), and Scots pine (*Pinus sylvestris* L.) are the dominant tree species. Over the past two decades, a decline in spruce stands has been observed due to the combined effects of drought, the fungus *Armillaria* spp., and a bark beetle infestation. In the survey area, adults of *I. duplicatus* were collected in 2020 from which the fungus *Beauveria* was isolated and experiments with infected pheromone traps were established in 2022.

2.2. Insect sampling for screening of Beauveria spp.

The screening for *Beauveria* species within natural populations of *I. duplicatus* was carried out during October and November 2020. There were two types of samples collected: spruce bark and needle litter. Bark samples were taken from ten freshly cut spruce trees that were infested with bark beetles. The samples were from two distinct locations and a total of 30 bark pieces (approximately 30 cm × 20 cm) were taken. The samples of needle litter were gathered from the forest floor in three different places. A set of 30 samples was taken from under 30 spruces infested by bark beetles. A typical litter sample weighed 500–1000 g and had a volume of 1200–3500 ml. The litter sample (20 cm × 20 cm × 10 cm; l × w × h) was collected on the base of the tree trunk from the southern side.

The bark and litter samples were stored in polyethylene bags in a refrigerator for four weeks. Bark beetles were selected from the samples with a magnifying glass and it was visually done based on macroscopic symptoms for individuals infected with *Beauveria*. The infected adults were placed separately into storage in 2 mL plastic microtubes at 8 °C before examining under a dissecting stereomicroscope (50 ×) and confirming mycosis by *Beauveria*. Imagines with the disease were used for obtaining *in vitro* cultures.

2.3. Isolation and cultivation of Beauveria spp.

Sabouraud dextrose agar (SDA) plates supplemented with fungicides (500 mg L^{-1} dodine and 250 mg L^{-1} cycloheximide) and antibiotics (50 mg L^{-1} tetracycline hydrochloride and 600 mg L^{-1} streptomycin sulfate) were inoculated with spores produced externally on the surface of infected beetles. They were then incubated at 25 °C ± 2°C for 14 days and developing fungal colonies were transferred onto fresh SDA plates with no fungicides and antibiotics. These colonies were maintained at 25 °C ± 2°C or onto SDA slants and stored at 4 °C. All *in vitro* cultures were deposited in the fungal collection at the Institute of Forest Ecology at the Slovak Academy of Sciences (Nitra, Slovakia).

Spores from 10-day-old fungal cultures, produced on SDA plates at 25 °C ± 2°C, were used in the bioassays of this study. Stocks of spore suspensions were prepared by putting them in 100 ml of 0.05 % (w/v) Tween® 80. The suspensions were vortexed for 60 s and mycelial fragments with spore clusters were eliminated by filtration through a sterile 10-μm nylon membrane (Spectra Mesh®, Spectrum Laboratories, Inc., USA). Using a Neubauer hemocytometer, the concentration of spores in stock suspensions was determined, and the one required for bioassays was obtained by diluting these with 0.05 % (w/v) Tween® 80. The percentage of viable spores in suspension was assessed by doing germination tests on SDA plates at 25 °C ± 2 °C before each bioassay. Depending on the strains, it ranged between 96.30 % ± 0.93 % and 98.90 % ± 0.50 % evaluated after 12 h of incubation.

2.4. Identification of isolated fungi

In vitro cultures were tentatively identified by a microscopic (500 ×) examination according to the morphology of fungal microstructures (Rehner and Buckley, 2005; Rehner et al., 2011; Humber, 2012; Khonsanit et al., 2020). The morphological identification was supported by determining the internal transcribed spacer (ITS) region of the ribosomal DNA (ITS1-5.8S-ITS2) and a partial sequence of the translation elongation factor 1-alpha (TEF1-α) gene. DNA was extracted from fungal biomass of *in vitro* grown cultures using EZ-10 Spin Column Fungal Genomic DNA Miniprep Kit (Bio Basic Canada Inc., Ontario, Canada) according to the manufacturer's protocol. Extracted DNA was suspended in 50 μL of the elution buffer and stored at −20 °C. ITS region and TEF1-α gene were amplified with the primer pairs ITS1-F/ITS4 (White et al., 1990; Gardes and Bruns, 1993) and EF-983F/EF-2218R (Rehner and Buckley, 2005), respectively. The PCR reaction mixture (20 μL) contained 5 × Hot FIREPol® Blend Master Mix (Solis BioDyne OÜ, Tartu, Estonia), 0.2 μM of each primer, deionized water for molecular biology grade and 50 ng of template DNA. PCRs were run in a Bio-Rad T100^{TM} Thermal Cycler (Bio-Rad Laboratories Inc., CA, USA) according to Barta et al. (2020). The resulting PCR products were separated by electrophoresis on 1 % agarose gel using SimplyBlue^{TM} SafeStain (EURx Ltd., Gdańsk, Poland) and visualized under UV light. The target PCR fragments were purified by QIAquick PCR Purification Kit (Qiagen N.V., Venlo, The Netherlands) and sequenced by the Sanger method through the use of Macrogen DNA Sequencing Service (Macrogen Europe B.V., Amsterdam, The Netherlands). Acquired sequences were processed in SnapGene® Viewer version 6.2.1 (GSL Biotech LLC, CA, USA) and compared by BLASTn (Zhang et al., 2000) against DNA data deposited in the NCBI GenBank Sequence Database.

2.5. In vitro extracellular enzyme activity of Beauveria bassiana strains

The pH indicator, bromocresol purple, in agar media was used for the detection of extracellular enzymes produced by strains *in vitro*. Protease activity was tested in agar medium (0.01 % yeast extract, 2 % agar, 0.01 % bromocresol purple, adjusted to pH 6) with 1 % skimmed milk and the chitinase activity was measured in chitin-yeast extract-agar medium (2 % colloidal chitin, 0.05 % yeast extract, 2 % agar, 0.01 % bromocresol purple, pH at 6.6) (Bai et al., 2012). The lipase activity of strains was evaluated on an agar medium (Padmini and Padmaja, 2013) consisting of peptone (1.5 %), NaCl (0.5 %), $CaCl_2$ (0.1 %), Tween 20 (1 %), agar (2 %) and bromocresol purple (0.01 %), pH was adjusted to 6.6.

Autoclaved media were poured into sterilized Petri dishes (90 mm, diameter) and allowed to solidify. Using a heated glass rod (5 mm in diameter), nine wells (approximately 3 mm in depth) were created on each agar plate's surface. Subsequently, 10 μL of the suspension from tested *Beauveria* strains (a concentration of 1 × 10^6 spores mL^{-1}) were pipetted in wells and incubated at 25 °C ± 2 °C for 96 h. The amount of enzyme production was determined based on the difference in diameter of the yellow halo zone formed on media around the colony in culture. The diameters of fungal colonies and halo zones in the vicinity were

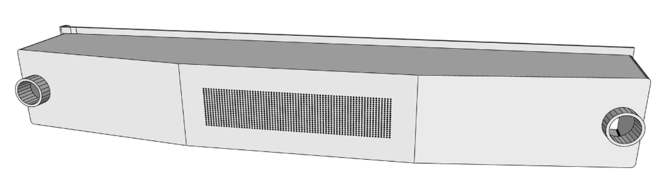

Fig. 1. A trap applicator of fungal inoculum is made from transparent plastic material and inside there are holders for storing carriers (bottom view).

Fig. 2. A pheromone trap equipped with an applicator of fungal inoculum and plastic bottles for one day capturing infected/control beetles.

measured. The enzymatic activity (*EA*) was calculated depending on the diameter of a halo zone that surrounds a fungal colony, divided by the diameter of the colony. Each agar plate that had nine inoculation points served as a replicate and five plates were done per treatment and strain. The enzymatic activity index (*EAI*) was calculated as follows:

$$EAI = \frac{mea}{\left(\frac{\sum (ea - mea)^2}{3}\right) + 1}$$

where *ea* is the value for the activity of extracellular enzymes (proteases, chitinases, or lipases) calculated based on the diameter of a halo zone and *mea* is the mean enzymatic activity according to the values of all extracellular enzymes tested. Six strains with the highest *EAI* (> 2.00) were selected for the pathogenicity testing.

2.6. Pathogenicity tests for Beauveria bassiana strains

Adult bark beetles for the pathogenicity tests were collected with traps baited with ID Ecolure (Fytofarm s.r.o., Slovakia). We used one-day trapping. The adults were kept in plastic containers with ventilation holes and pieces of spruce bark in a refrigerator (8 °C) until use, which was usually no longer than two days.

Six *Beauveria* strains with the highest enzymatic activity were chosen for pathogenicity tests against *I. duplicatus* adults. As a reference, the Slovak strain of *B. bassiana* DSM 32081 (ARSEF 12957) was tested. It was isolated from *I. typographus*, showed high virulence against adult spruce bark beetles (Barta et al., 2018a), and was selected as a promising bioagent against this pest (Barta et al., 2018b).

To evaluate the pathogenicity of the fungal strains, adults were submerged in the suspensions with a concentration of 1×10^6 spores mL^{-1}. For each strain, groups of 33 individuals were treated by submerging them for 5 s. The treated individuals were then placed into Petri dishes (60 mm × 15 mm diameter) and incubated for 15 days at 25 °C ± 2 °C, saturated RH, and a 12/12 h (L/D) light regime. Three treated bark beetles were incubated together in a Petri dish with moistened cellulose and a piece of fresh bark (3 cm × 3 cm) as food. A group of 33 adults was immersed in sterile 0.01 % Tween® 80 as controls. The dishes were monitored at 24-hour intervals for 15 days. Adults with the symptoms of *Beauveria* infection were removed from the Petri dish to avoid further contamination of the others. The cadavers were microscopically examined, and only those being infected were used to calculate pathogenicity. Each strain was tested three times.

2.7. Effect of selected spruce monoterpenes on growing Beauveria bassiana strains

Monoterpenes, such as (−)-α-pinene, (+)-α-pinene, (−)-β-pinene, (S)-(−)-limonene, and (R)-(+)-limonene (Sigma-Aldrich®, USA) were tested for their effects on the growth of *B. bassiana* strains on agar media. SDA medium was enriched with monoterpenes at a concentration of 1 % (v/v) before autoclaving at 121 °C, 210 kPa for 20 min. Dimethyl sulfoxide (1 %; v/v) was added to enhance the solubility of monoterpenes before autoclave use and sterilized media were dispensed into Petri dishes (20 mL per 90 mm dish) with Systec MediaPrep (Systec GmbH & Co. KG, Linden, Germany). The experimental setup involved the placement of four autoclaved filter paper discs (6 mm in diameter) on the surface of the solidified agar medium in the dishes. The discs were evenly arranged in a square formation, with a distance of 35 mm between each.

Strains from the pathogenicity tests were included in this bioassay. Spore suspensions (1×10^8 spores mL^{-1}) from each strain were prepared as above and 10 μL were pipetted to the center of any filter paper disc. Each monoterpene was tested using five SDA plates with paper discs. A set of five plates were treated with 0.05 % (w/v) Tween® 80 and served as control samples. The SDA plates were subjected to seven days of incubation at 25 °C ± 2 °C and a natural photoperiod. Following a 7-day period, the diameter of fungal colonies was measured with a digital caliper. Diameter measurement for every colony was conducted at the narrowest and widest points, then determining the mean diameter. The inhibitory effect of monoterpenes on the radial growth (*EMRG*) for fungal colonies was calculated as follows:

$$EMGR = \frac{(Dc - Dm)}{Dc} \times 100[\%]$$

where *Dc* is the mean diameter of a fungal colony in the control and *Dm* is the mean diameter of a fungal colony treated with monoterpenes.

2.8. Field infection of Ips duplicatus in pheromone traps

On May 19, 2022, a total of eight adjusted MultiWit Bark Beetle Slit Traps (Witasek PflanzenSchutz GmbH) were deployed within two areas at the research site. Three traps with *B. bassiana* strain NRID11 and one trap as the control were positioned in area A, and three traps with *B. bassiana* strain NRID43 and one trap as the control were in area B, which was in a 100 m distance from area A. ID Ecolure lures (Fytofarm s. r.o.) were used in all traps to attract *I. duplicatus*.

A trap applicator of fungal inocula (Fig. 1), for which a registered community design was submitted in 2023 (No. 015023460–0001), was loaded with a carrier of the *B. bassiana* inoculum. The applicator is a modified trap collection container that protects the carrier from water and high temperature, ensures contact of trapped beetles with spores of entomopathogen, and allows an escape of infected beetles into the environment. The carrier of a biologically active organism (European patent No. EP 3836789 from 2024) had a diameter of about 25 mm and was produced from wheat flour, cereal brands, and egg on which the entomopathogenic fungus could grow and sporulate (Lalík et al., 2021). In each trap applicator, two carriers overgrown with a particular *B. bassiana* strain were used. Strains in use in field experiments were selected based on the results of pathogenicity and resistance tests against monoterpenes. The control traps did not contain the inoculated carrier.

During swarming, beetles were captured on four separate days using pheromone traps. We investigated the impact of exposure duration in the field for *B. bassiana* strains in carriers on the pathogenic infection efficiency in beetles. These carriers were evaluated over a 60-day period cycle, with beetles collected in four phases: A (3 days), B (21 days), C (30 days), and D (60 days). Each trap yielded at least 25 beetles per collection period.

Lures were inserted into the traps in the morning of the catching day.

Table 1
List of *Beauveria bassiana* strains obtained from adults of *Ips duplicatus* collected from either spruce bark or litter samples in northwestern Slovakia.

Strain code	Source of sample	GenBank accession no.[a]		Extracellular enzymatic activity ± SE[b]			Enzymatic activity index
		ITS	TEF-1α	PA	CA	LA	
NRID01	Spruce bark	OQ911944	OQ927328	2.30 ± 0.13	1.35 ± 0.04	2.74 ± 0.11	1.59
NRID02	**Spruce bark**	**OQ911945**	**OQ927329**	**2.30 ± 0.04**	**3.37 ± 0.06**	**2.92 ± 0.14**	**2.35**
NRID04	Spruce bark	OQ911946	OQ927330	1.83 ± 0.09	2.45 ± 0.03	1.95 ± 0.07	1.94
NRID05	Spruce bark	OQ911947	OQ927331	1.48 ± 0.04	2.72 ± 0.11	2.12 ± 0.08	1.68
NRID06	Spruce bark	OQ911948	OQ927332	1.98 ± 0.06	1.95 ± 0.08	2.01 ± 0.11	1.98
NRID07	Spruce bark	OQ911949	OQ927333	1.39 ± 0.07	2.66 ± 0.09	2.41 ± 0.11	1.65
NRID08	**Spruce bark**	**OQ911950**	**OQ927334**	**2.25 ± 0.05**	**2.91 ± 0.15**	**2.38 ± 0.11**	**2.33**
NRID09	**Spruce bark**	**OQ911951**	**OQ927335**	**1.88 ± 0.07**	**3.75 ± 0.12**	**2.75 ± 0.10**	**1.76**
NRID10	Spruce bark	OQ911952	OQ927336	2.02 ± 0.06	1.91 ± 0.13	2.82 ± 0.15	1.94
NRID11	**Spruce bark**	**OQ911953**	**OQ927337**	**2.30 ± 0.09**	**3.22 ± 0.12**	**2.55 ± 0.12**	**2.33**
NRID12	Spruce bark	OQ911954	OQ927338	1.74 ± 0.12	1.91 ± 0.10	2.72 ± 0.12	1.80
NRID13	Spruce bark	OQ911955	OQ927339	1.77 ± 0.10	2.81 ± 0.06	2.19 ± 0.07	1.91
NRID14	Spruce bark	OQ911956	OQ927340	1.91 ± 0.12	1.99 ± 0.08	1.85 ± 0.14	1.91
NRID15	Spruce bark	OQ911957	OQ927341	1.50 ± 0.04	1.48 ± 0.05	2.02 ± 0.10	1.57
NRID17	Spruce bark	OQ911958	OQ927342	1.64 ± 0.07	2.46 ± 0.06	2.00 ± 0.06	1.83
NRID18	Spruce bark	OQ911959	OQ927343	1.58 ± 0.07	2.51 ± 0.09	2.44 ± 0.10	1.85
NRID36	Spruce bark	OQ911960	OQ927344	1.07 ± 0.04	1.39 ± 0.06	2.29 ± 0.12	1.25
NRID37	Spruce bark	OQ911961	OQ927345	1.66 ± 0.06	2.61 ± 0.06	2.02 ± 0.12	1.82
NRID38	Spruce bark	OQ911962	OQ927346	1.68 ± 0.09	2.44 ± 0.12	2.24 ± 0.17	1.92
NRID39	Spruce bark	OQ911963	OQ927347	1.70 ± 0.04	2.75 ± 0.12	2.20 ± 0.10	1.87
NRID40	Spruce bark	OQ911964	OQ927348	1.64 ± 0.09	2.24 ± 0.09	2.36 ± 0.10	1.89
NRID41	Spruce bark	OQ911965	OQ927349	1.00 ± 0.01	1.00 ± 0.01	2.07 ± 0.11	1.08
NRID42	Spruce bark	OQ911966	OQ927350	1.98 ± 0.12	2.10 ± 0.09	1.80 ± 0.08	1.93
NRID43	**Spruce bark**	**OQ911967**	**OQ927351**	**2.30 ± 0.06**	**2.26 ± 0.06**	**2.84 ± 0.18**	**2.31**
NRID47	Spruce bark	OQ911968	OQ927352	1.93 ± 0.04	1.76 ± 0.13	2.55 ± 0.11	1.87
NRID48	Spruce bark	OQ911969	OQ927353	1.87 ± 0.09	1.24 ± 0.06	2.36 ± 0.13	1.51
NRID49	Spruce bark	OQ911970	OQ927354	1.79 ± 0.07	2.19 ± 0.09	2.89 ± 0.08	1.89
NRID50	Spruce bark	OQ911971	OQ927355	1.84 ± 0.04	1.27 ± 0.04	3.05 ± 0.06	1.32
NRID51	Spruce bark	OQ911972	OQ927356	1.70 ± 0.05	1.32 ± 0.08	2.66 ± 0.09	1.44
NRID52	Spruce bark	OQ911973	OQ927357	1.60 ± 0.06	2.68 ± 0.08	2.32 ± 0.12	1.83
NRID53	Spruce bark	OQ911974	OQ927358	1.87 ± 0.10	1.44 ± 0.07	2.65 ± 0.13	1.59
NRID55	Spruce bark	OQ911975	OQ927359	2.18 ± 0.12	1.34 ± 0.12	2.53 ± 0.21	1.61
NRID58	Spruce bark	OQ911976	OQ927360	1.96 ± 0.06	2.10 ± 0.07	1.94 ± 0.12	1.99
NRID61	Spruce bark	OQ911977	OQ927361	2.02 ± 0.04	1.80 ± 0.15	3.07 ± 0.13	1.75
NRID64	Spruce bark	OQ911978	OQ927362	2.10 ± 0.16	1.46 ± 0.08	2.64 ± 0.12	1.67
NRID65	Spruce bark	OQ911979	OQ927363	2.05 ± 0.05	1.49 ± 0.06	2.73 ± 0.18	1.66
NRID66	Spruce bark	OQ911980	OQ927364	1.00 ± 0.01	1.25 ± 0.08	1.57 ± 0.13	1.21
NRID67	Spruce bark	OQ911981	OQ927365	2.08 ± 0.08	1.44 ± 0.03	2.51 ± 0.10	1.68
NRID68	Spruce bark	OQ911982	OQ927366	1.22 ± 0.09	1.00 ± 0.12	2.56 ± 0.08	1.08
NRID69	Spruce bark	OQ911983	OQ927367	1.30 ± 0.05	1.52 ± 0.01	2.66 ± 0.08	1.35
NRID71	Spruce bark	OQ911984	OQ927368	1.70 ± 0.06	2.71 ± 0.05	1.90 ± 0.14	1.77
NRID72	Spruce bark	OQ911985	OQ927369	1.46 ± 0.12	2.70 ± 0.16	2.31 ± 0.11	1.70
NRID73	Spruce bark	OQ911986	OQ927370	1.42 ± 0.11	1.88 ± 0.09	2.34 ± 0.05	1.65
NRID74	Spruce bark	OQ911987	OQ927371	1.84 ± 0.07	1.98 ± 0.05	2.31 ± 0.07	1.97
NRID75	Spruce bark	OQ911988	OQ927372	1.99 ± 0.10	1.92 ± 0.09	2.10 ± 0.08	1.99
NRID76	Spruce bark	OQ911989	OQ927373	1.67 ± 0.06	2.92 ± 0.15	2.34 ± 0.13	1.83
NRID77	Spruce bark	OQ911990	OQ927374	1.92 ± 0.04	1.75 ± 0.10	2.59 ± 0.08	1.84
NRID78	**Needle litter**	**OQ911991**	**OQ927375**	**2.12 ± 0.11**	**2.22 ± 0.19**	**2.79 ± 0.12**	**2.22**

B. bassiana strains highlighted in bold letters were selected for pathogenicity tests.
[a] Accession numbers for internal transcribed spacer (ITS) and translation elongation factor 1-alpha (TEF-1α) sequences deposited in the GenBank database;
[b] Mean enzymatic activity (PA – protease activity, CA – chitinase activity, LA – lipase activity) of strains *in vitro*.

Table 2
Inhibitory effects of monoterpenes on the radial growth of *Beauveria bassiana* strains under laboratory conditions.

Strains	Inhibitory effects of monoterpenes ± SE (%)[a]				
	(−)-α-pinene	(+)-α-pinene	(−)-β-pinene	(S)-(−)-limonene	(R)-(+)-limonene
NRID02	11.97 ± 1.66 ab	100.00 a	25.10 ± 0.71 a	100.00	100.00
NRID08	8.84 ± 1.72b	100.00 a	9.44 ± 2.31b	100.00	100.00
NRID09	18.97 ± 3.58 bc	100.00 a	18.97 ± 3.58b	100.00	100.00
NRID11	−3.18 ± 0.70 d	11.51 ± 0.57b	7.93 ± 1.00b	100.00	100.00
NRID43	−25.45 ± 1.54 e	81.36 ± 3.49c	−2.80 ± 1.65c	100.00	100.00
NRID78	9.48 ± 0.64 bf	70.82 ± 2.48c	16.40 ± 0.86 d	100.00	100.00
DSM 32,081	4.43 ± 0.52 g	58.15 ± 1.17 d	18.36 ± 0.31 e	100.00	100.00
Kruskal-Wallis test	$\chi^2 = 107.91$; $p < 0.001$	$\chi^2 = 108.19$; $p < 0.001$	$\chi^2 = 96.89$; $p < 0.001$	–	–

[a] Values followed with the same letter in the column are not significantly different ($p = 0.05$) according to the pairwise Mann-Whitney test.

Table 3
Cumulative daily mortality rate (%) and mean survival time (MST) of *Ips duplicatus* treated with *Beauveria bassiana* strains under laboratory conditions.

DAT[a]	Cumulative daily mortality rate (%) ± SE for *Beauveria bassiana* strains							
	NRID02	NRID08	NRID09	NRID11	NRID43	NRID78	DSM 32,081	ANOVA[b]
4	0.00	0.00	0.00	0.00	0.00	4.04 ± 2.02 a	13.13 ± 3.64b	$F_{(6,14)} = 13.55$, $p < 0.001$
5	37.50 ± 5.41 ab	40.63 ± 4.77 ab	30.21 ± 9.26 ab	42.71 ± 7.29b	14.58 ± 2.08 a	14.04 ± 2.02 a	43.75 ± 3.13b	$F_{(6,14)} = 12.44$, $p < 0.001$
6	64.21 ± 9.36 ab	60.00 ± 9.36 ab	51.58 ± 7.37 ab	72.63 ± 4.21 ab	40.00 ± 3.65 a	77.89 ± 1.82b	57.89 ± 8.22 ab	$F_{(6,14)} = 3.44$, $p = 0.027$
7	86.32 ± 4.21 ab	81.05 ± 5.47 ab	69.47 ± 5.86 a	94.74 ± 1.05b	81.05 ± 6.57 ab	77.89 ± 1.82 ab	77.63 ± 6.40 ab	$F_{(6,14)} = 3.40$, $p = 0.028$
8	91.58 ± 2.11 a	90.53 ± 1.82 a	81.05 ± 4.82 a	97.89 ± 1.05 a	90.53 ± 4.82 a	82.11 ± 1.05 a	86.32 ± 5.57 a	$F_{(6,14)} = 2.52$, $p = 0.072$
9	93.68 ± 0.00 ab	93.68 ± 1.82 ab	83.16 ± 4.21 a	98.95 ± 1.05b	93.68 ± 3.64 ab	82.11 ± 1.05 a	95.79 ± 2.11 ab	$F_{(6,14)} = 4.65$, $p = 0.008$
10	93.68 ± 0.00 ab	95.79 ± 1.05 abc	85.26 ± 4.59 a	100.00c	97.89 ± 1.05 bc	84.21 ± 1.82 a	97.89 ± 2.11 bc	$F_{(6,14)} = 9.70$, $p < 0.001$
11	94.74 ± 1.05 ab	95.79 ± 1.05 abc	88.42 ± 2.79 a	100.00c	98.95 ± 1.05 bc	84.21 ± 1.82 a	97.89 ± 2.11 bc	$F_{(6,14)} = 10.81$, $p < 0.001$
12	94.74 ± 1.05 abc	97.89 ± 1.05 bc	91.58 ± 2.11 ab	100.00c	98.95 ± 1.05 bc	84.21 ± 1.82 a	97.89 ± 2.11 bc	$F_{(6,14)} = 8.68$, $p < 0.001$
13	94.74 ± 1.05 ab	97.89 ± 1.05b	94.74 ± 1.05 ab	100.00b	98.95 ± 1.05b	84.21 ± 1.82 a	97.89 ± 2.11b	$F_{(6,14)} = 8.21$, $p < 0.001$
MST[c]	6.59 ± 0.27	6.54 ± 0.24	7.32 ± 0.31	5.90 ± 0.10	6.12 ± 0.17	7.18 ± 0.28	6.43 ± 0.25	

[a] DAT – Days after treatment of *I. duplicatus* with conidia from *B. bassiana* strains;
[b] ANOVA – Parameters for analysis of variance, testing the differences in mortality data for *I. duplicatus* treated with tested *B. bassiana* strains; mean values in the row followed with the same letter are not varying significantly ($p = 0.05$; Tukey's HSD test);
[c] MST – Mean survival time (± SE) (in days) of *I. duplicatus* treated with tested *B. bassiana* strains estimated by the Kaplan-Meier's analysis.

The beetles were captured into vented plastic bottles attached to both sides of the collection container (Fig. 2). Afterwards, these beetles were taken to the laboratory and reared in glass Petri dishes (90 mm in diameter) for 12 days. Each dish contained moistened cellulose and two pieces of fresh spruce bark (3 cm × 3 cm). Five beetles were placed into each Petri dish. Mortality and signs of *B. bassiana* infection were monitored at one-day intervals. All cadavers were microscopically examined, and only those being infected were used to calculate pathogenicity. Beetles with the symptoms of *Beauveria* infection were removed from the Petri dish to avoid further contamination of the others.

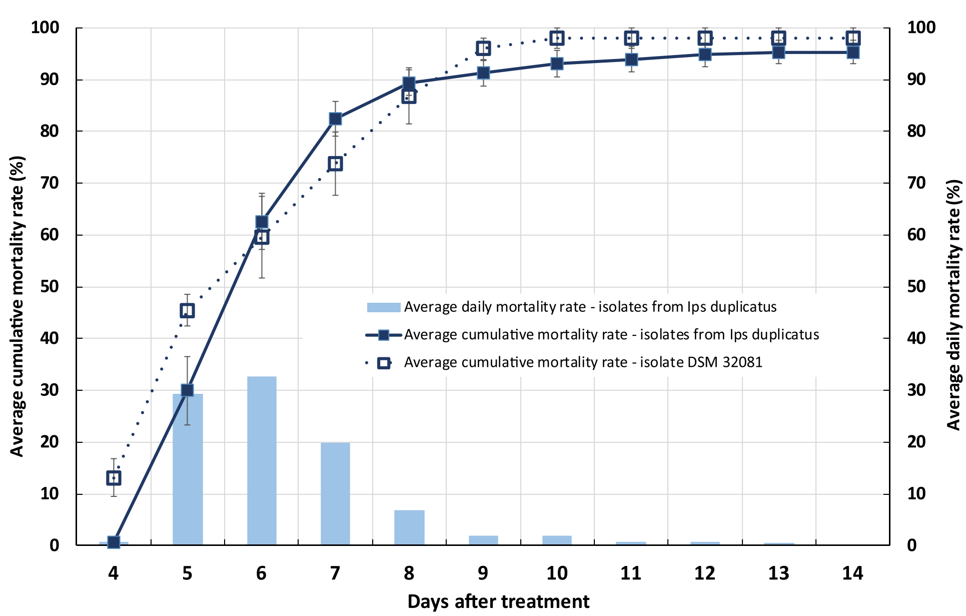

Fig. 3. Average cumulative mortality and mean daily mortality rates of *Ips duplicatus* treated with spores of *Beauveria bassiana* (1×10^6 spores mL^{-1}) in laboratory tests for pathogenicity calculated from pooled data of strains obtained in this study. The error bars represent the standard errors of the means. The mean mortality in the control reached 4.04 ± 2.02 %.

2.9. Data analyses

The mortality of bark beetles treated with *B. bassiana* strains in laboratory tests was adjusted for control mortality (Schneider-Orelli's correction). The corrected mortality rates were evaluated for normality (Shapiro-Wilk's test) and homoscedasticity (Bartlett's test) and then an arcsine transformation ($n' = \arcsin\sqrt{n}$) was performed to obtain a normally distributed dataset before being subjected to ANOVA. If significant differences ($p = 0.05$) were detected, the post-hoc Tukey's HSD test was used to separate and compare means. Mean survival times of bark beetles treated with strains of *B. bassiana* were estimated with Kaplan-Meier's analysis. The log-rank test was performed to challenge the null hypothesis of no difference in survival between bark beetles treated with various *B. bassiana* strains. Effects data for monoterpenes on the fungal growth was tested for normality and homoscedasticity. The Kruskal-Wallis test was done to compare what monoterpenes did among the fungal inocula and it was followed by the pairwise Mann-Whitney tests for analyzing specific pairs of *B. bassiana* strains.

The fungal infection of bark beetles in pheromone traps was assessed on the 6th and 12th days of the field testing, and these timeframes were used for statistical analysis. The effects of the variants (NRID11, NRID43, and control) on mortality have been investigated by employing a generalized linear model (GLM) with a binomial distribution and a logit link function. The model was fit to fungal infection (infected, non-infected) as a response and process variable (NRID11, NRID43, control) and to the phase where the carrier is exposed in traps (A – 3 days, B – 21 days, C – 30 days, D – 60 days) as a categorical explanatory variable. The effect of variants was compared with the Tukey method for benchmarking family estimates. All analyses were performed using R version 4.2.3 (2023–03-15 ucrt) and RStudio 2023.03.0 + 386 (R Core Team, 2023). Data manipulation was done with the dplyr package (Wickham et al., 2023). Estimated marginal means were calculated with the emmeans package (Lenth, 2023).

3. Results

3.1. Entomopathogenic fungi isolated from Ips duplicatus

Adults of *I. duplicatus* were found to be susceptible to *Beauveria* infection. Fungal disease symptoms were observed in 61 (27.1 %) of 225 beetles collected from 30 bark samples and 47 strains were obtained *in vitro* (Table 1). Only 15 individuals of *I. duplicatus* were found in needle litter samples, and five of them were exhibiting symptoms of an infection with *Beauveria* infection. In the needle litter, 83 individuals of *I. typographus* were also discovered in total, with 24 of them displaying signs of *Beauveria* infection. Additionally, one individual of the species *Pityogenes chalcographus* (L.) was found to be infected with *Beauveria* sp. From the forest floor sampling, a single strain of *B. bassiana* (NRID78) was obtained. Altogether, 48 strains were found *in vitro* from *I. duplicatus* and all were identified as *B. bassiana* by determining ITS and TEF-1α DNA regions. The comparison of the sequences with GenBank data indicated a high level of similarity (around 99 %) with the neotype *B. bassiana* strain (ARSEF 1564). The sequences gained from this study were deposited in the GenBank database and their accession numbers are listed in Table 1. *Beauveria bassiana* cultures were screened for enzymatic activity, and subsequently selected strains were evaluated for pathogenicity against *I. duplicatus* adults.

Table 4
Results of the log-rank test comparing survival curves for *Ips duplicatus* adults inoculated with *Beauveria bassiana* strains in laboratory bioassays.

Strains	NRID08	NRID09	NRID11	NRID43	NRID78	DSM 32,081
NRID02	0.09, *p* = 0.768	2.29, *p* = 0.130	2.05, *p* = 0.152	1.38, *p* = 0.241	**5.69, *p* = 0.017**	0.06, *p* = 0.813
NRID08	–	2.03, *p* = 0.155	3.29, *p* = 0.070	0.55, *p* = 0.456	**7.30, *p* = 0.007**	0.04, *p* = 0.835
NRID09	–	–	**10.49, *p* = 0.001**	0.97, *p* = 0.324	2.31, *p* = 0.129	2.65, *p* = 0.104
NRID11	–	–	–	**9.90, *p* = 0.002**	**13.19, *p* = 0.001**	2.87, *p* = 0.090
NRID43	–	–	–	–	1.20, *p* = 0.273	0.53, *p* = 0.465
NRID78	–	–	–	–	–	**7.72, *p* = 0.005**

For data (χ^2 statistics and *p*-value) written in bold letters, we rejected the null hypothesis ($p < 0.05$) in which the survival rates of beetles treated with the two *Beauveria* strains were statistically equivalent.

3.2. Extracellular enzymatic activity of Beauveria bassiana strains

The results of the agar plate-based screening method used to assess the enzymatic activity of *B. bassiana* strains are shown in Table 1. All those that were tested exhibited enzyme production on agar plates, as evidenced by the formation of halos around the fungal cultures. The width of the halo zone served as an indicator of it, with a wider halo suggesting higher enzyme production. A significant difference ($p < 0.001$) in producing them was observed among fungal strains. Protease activity values varied significantly from 1.00 to 2.30 ($F_{(47, 384)} = 16.98$, $p < 0.001$), with strains NRID01, NRID02, NRID08, NRID11, and NRID43 exhibiting the highest ones. Chitinase activity also changed significantly among these ($F_{(47, 384)} = 46.55$, $p < 0.001$) and ranged from 1.00 to 3.75. Strains NRID02, NRID08, NRID09, NRID11, and NRID76 displayed the largest halos on chitin yeast extract agar, indicating the highest production. Values of lipolytic activity were from 1.57 to 3.07, and the differences between strains were also significant ($F_{(47, 384)} = 7.32$, $p < 0.001$). The strains NRID50, NRID61, NRID02, NRID09, and NRID49 had the highest. The extracellular enzymatic activity of entomopathogenic fungi plays a crucial role in the pathogenesis process. Hydrolytic enzymes, such as chitinases, proteases, and lipases, are responsible for degrading the insect's integument, enabling the fungus to penetrate and infect the host. Strains of *B. bassiana* with the highest enzymatic activity (Table 1) were chosen for the pathogenicity tests in this study. Based on the enzymatic activity indices, the following *B. bassiana* strains were selected ($EAI > 2.00$): NRID02, NRID08, NRID09, NRID11, NRID43, and NRID78.

3.3. Effect of selected spruce monoterpenes on growing Beauveria bassiana strains

The data in Table 2 shows the impact of monoterpenes on the radial growth of *B. bassiana* strains. The inhibitory effects varied depending on specific monoterpenes and fungal strains. For example, 1 % limonene completely suppressed its growth across all strains, while the effects of α-pinene and β-pinene differed significantly ($p < 0.001$) among them. There was considerable variability in the impact on growth of *B. bassiana* for (−)-α-pinene and (−)-β-pinene, with the inhibitory effect reaching up to 25 %. However, two strains responded positively to these monoterpenes, NRID11 showing a 3 % increase in growth when (−)-α-pinene was present, and NRID43 exhibiting a 2.8 % or 25 % rise when (−)-β-pinene or (−)-α-pinene, respectively, were added to the agar medium. Furthermore (+)-α-pinene had a significantly higher ($p < 0.01$) inhibitory effect on the radial growth compared to the previously mentioned monoterpenes, reducing it by 58–100 % depending on the strain.

3.4. Pathogenicity tests for selected Beauveria bassiana strains

All strains tested were pathogenic to adult bark beetles and significantly ($F_{(7,16)} = 71.64$, $p < 0.001$) reduced their viability compared to the control group. The mean mortality of these beetles at the end of the bioassay (14 days after treatment) varied between 84 and 100 %

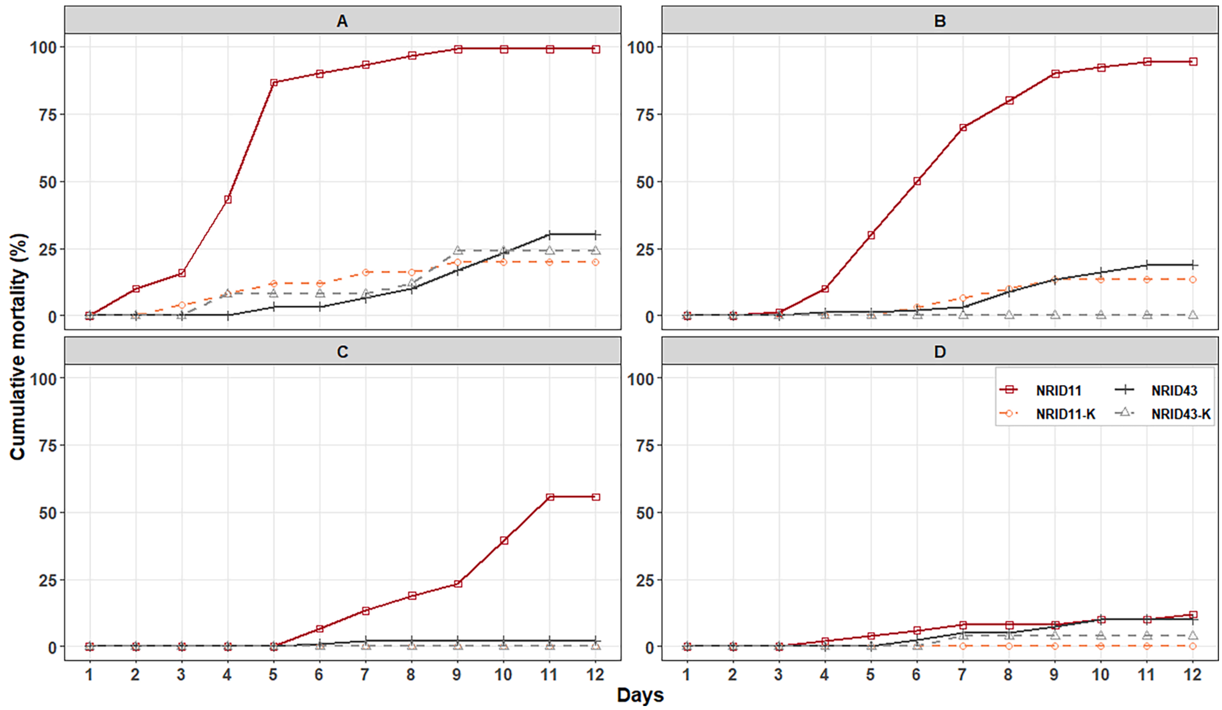

Fig. 4. Daily mortality rate of *I. duplicatus* infected by *B. bassiana* spores through carriers in pheromone traps. The pathogenicity was studied under laboratory conditions. This figure displays the cumulative mortality rates (%) of *I. duplicatus* subjected to *B. bassiana* strains NRID11 and NRID43 in pheromone traps using a carrier as an applicator. The subfigures (A, B, C, and D) represent the duration of the carriers in the traps (3, 21, 30, and 60 days), with D reflecting the longest deployment. The trend observed from A to D indicates that their prolonged exposure in pheromone traps leads to a decrease in the impact of the pathogen, emphasizing the importance of optimizing the length of time they are exposed to maintain the effectiveness of biocontrol agents.

Table 5
Estimated parameters in assessing the effects of carriers on the mortality of beetles after six days in Petri dishes.

Variable	Estimates	2.50 %	97.50 %	SE	Z-score	P-value	
(intercept) – control	−2.18	−3.01	−1.49	0.38	−5.69	< 0.001	***
NRID11	3.88	3.15	4.75	0.41	9.58	< 0.001	***
NRID43	0.10	−0.94	1.14	0.52	0.19	0.85	
phase – B	−0.95	−1.52	−0.39	0.29	−3.28	< 0.001	***
phase – C	−3.45	−4.20	−2.76	0.37	−9.41	< 0.001	***
phase – D	−3.57	−4.56	−2.72	0.46	−7.71	< 0.001	***

depending on the strain (Table 3). Adults that died during pathogenicity tests due to *Beauveria* infection displayed typical symptoms of white muscardine within about 24 h after the host's death. White mycelium grew out of the intersegmental membranes of cadavers, followed by the production of a thick white layer of sporulating mycelium covering nearly their entire surface. A general pattern of fungal disease development in the treated groups of bark beetles is shown in Fig. 3. The first dead adults with confirmed fungal infection occurred four days after the treatment when the average mortality for all strains from *I. duplicatus* in this study reached 0.67 %. On that day, only two strains (NRID78 and DSM 32081) killed the beetles. However, mortality due to fungal infection was recorded in all tested strains on day 5. On this particular day, the cumulative mortality significantly increased ($F_{(1,12)} = 30.17$, $p < 0.001$) to 29.97 ± 6.57 %. That is the cumulative mean number of deaths for strains obtained from *I. duplicatus*. The cumulative mortality induced by the reference strain DSM 32081 reached 45.45 ± 3.03 % on that day. The greatest increase in the daily mortality rate was observed between the fourth and fifth days, the mean value rose significantly ($F_{(1,12)} = 22.74$, $p < 0.001$) from 0.67 % to 29.29 %, followed by a constant increase on day 6 (32.66 ± 5.43 %), a moderate decrease on day 7 (19.87 ± 3.31 %), and a distinct reduction on days of rest during the assay. From the ninth day after treatment, the rise in daily mortality rate dropped below 2 %. On the 13th day, only 0.43 % of imagines died, and none were found on the next day when the bioassay was stopped. On its final day, the cumulative mortality ranged from 84.21 % to 100 %. Strain NRID11 was the only one that killed 100 % of tested beetles in the bioassays. The lowest mortality rate (84.21 %) was seen for strain NRID78. The pathogenicity of tested strains represented by the average cumulative mortality rate is shown in Table 3. The values varied significantly ($p < 0.05$) among all. The mean mortality in the control group reached 4.04 ± 2.02 %.

The average survival probability of beetles treated with *B. bassiana* strains, as estimated by the Kaplan-Meier analysis, decreased over time following the treatment. The log-rank test comparing survival curves for bark beetles inoculated with *B. bassiana* strains (Table 4) revealed

Table 6
Comparison of infection between variants/treatments and the time of carrier exposure after six days in Petri dishes based on the Tukey method for benchmarking family estimates. The log odds were exponentiated to obtain odds ratios (OR).

Contrast	OR	2.50 %	97.50 %	SE	P-value	
NRID11/control	50	0.01	0.05	0.01	< 0.001	***
NRID43/control	1.09	0.27	3.09	0.47	0.981	
NRID11/NRID43	44.02	17.28	112.15	17.56	< 0.001	***
A/B	2.58	1.23	5.42	0.75	< 0.001	***
A/C	31.42	12.26	80.56	11.52	< 0.001	***
A/D	35.60	10.83	117.05	16.49	< 0.001	***
B/C	12.18	4.90	30.30	4.32	< 0.001	***
B/D	13.80	4.30	44.34	6.27	< 0.001	***
C/D	1.13	0.32	4.07	0.56	0.994	

Table 7
Estimated parameters in assessing the effects of carriers on the mortality of beetles after 12 days in Petri dishes.

Variable	Estimates	2.50 %	97.50 %	SE	Z-score	P-value	
(intercept) – control	−1.35	−1.97	−0.80	0.30	−4.54	< 0.001	***
NRID11	4.90	4.16	5.73	0.40	12.32	< 0.001	***
NRID43	0.96	0.29	1.68	0.35	2.72	0.006	**
phase – B	−1.02	−1.66	−0.41	0.32	−3.23	0.001	**
phase – C	−3.35	−4.15	−2.63	0.39	−8.64	< 0.001	***
phase – D	−4.74	−5.72	−3.84	0.48	−9.94	< 0.001	***

significant ($p < 0.05$) variability among all. Strain NRID11, which was the most effective against these imagines in the bioassay, displayed a survival curve that was significantly different from those of NRID09, NRID43, and NRID78, but not from those of NRID02, NRID08, and DSM 32081. The estimated mean survival time (MST) of *I. duplicatus* after treatment varied from 5.90 to 7.32 days (Table 3), depending on the specific strains. Strain NRID11 had the shortest estimated value, while NRID09 had the longest. Based on the pathogenicity test, strain NRID11 was selected as the best. This is the only one that killed all bark beetles in the assay. Furthermore, the 100 % mortality was already recorded 10 days after the treatment. Strain NRID43 was found to be the second most effective, as it killed almost 99 % of the tested imagines within 11 days. These two strains were chosen for field testing.

3.5. *Infecting Ips duplicatus under field conditions using pheromone traps*

The first appearance of dead *I. duplicatus* individuals infected by strain NRID11 was observed within 6 days after a Petri dish placement during all experimental phases (A–D) (Fig. 4). In A, a 100 % infection rate was achieved within 9 days. In B, the progression was slower, and the rate of infection among imagines reached 92 %. This effectiveness diminished over time, as seen in subsequent periods. The trend from phases A to D indicates that prolonged exposure of the carriers within the pheromone traps leads to reduced efficacy of the inoculum.

We evaluated infected beetles on the 6th and 12th days after their placement into Petri dishes. For both the 6-day ($\chi^2_{(5)} = 457.85$, $p < 0.001$) and 12-day ($\chi^2_{(5)} = 584.68$, $p < 0.001$) periods, the fungal strain and the time of carrier exposure significantly influenced the likelihood of bark beetle infection.

For the 6-day period, there was a significant effect of treatment ($\chi^2_{(2)} = 284.79$, $p < 0.001$) and time of exposure ($\chi^2_{(3)} = 173.06$, $p < 0.001$) on the odds of infected beetles. Adults exposed to strain NRID11 had significantly higher estimates of infection compared to the control group (estimate = 3.88, $p < 0.001$) and NRID43 (estimate = 3.78, $p < 0.001$) (Table 5). When considering odds ratios, beetles exposed to NRID11 had much higher risks versus controls (odds ratio [OR] = 50, $p < 0.001$) and NRID43 (OR = 44.02, $p < 0.001$) (Table 6). There was no significant difference in infection probability between NRID43 and the control (OR = 1.09, $p = 0.981$). The duration of carrier exposure had a significant impact on infection efficiency, with phase A showing the highest values compared to phases B (OR = 2.58, $p < 0.001$), C (OR = 31.42, $p < 0.001$), and D (OR = 35.60, $p < 0.001$). Phase D had the lowest.

In the 12-day period, similar significant effects of treatment ($\chi^2_{(2)} = 383.62$, $p < 0.001$) and time of carrier exposure ($\chi^2_{(3)} = 201.06$, $p < 0.001$) on the likelihood of bark beetle infection were observed. Beetles exposed to strain NRID11 had significantly higher estimates of infection versus the control (estimate = 4.90, $p < 0.001$) and NRID43 (estimate = 3.94, $p < 0.001$) (Table 7). There was a significant difference between NRID43 and the control group (OR = 2.63, $p < 0.02$) (Table 8). The risks of infection for strain NRID11 remained significantly above both the control group (OR = 100, $p < 0.001$) and NRID43 (OR = 51.39, $p < 0.001$). The results for time of exposure were also similar, with beetles exposed in phase A showing much higher values compared to those in phases B (OR = 2.78, $p = 0.007$), C (OR = 28.57, $p < 0.001$), and D (OR = 114.32, $p < 0.001$).

4. Discussion

In our study, we obtained local strains of *B. bassiana* from naturally encountered populations of *I. duplicatus* and screened them for their enzymatic activity, sensitivity to monoterpenes, and pathogenicity against the species. Chosen strains were tested in modified pheromone traps with carriers under field conditions. Most published works on the entomopathogenic fungi tried against bark beetles are focused on commercial preparations containing *B. bassiana* (Kreutz et al., 2004a; Grodzki and Kosibowicz, 2015), only some deal with the testing of selected natural strains (Kreutz et al., 2004b; Mohan et al., 2007; Barta et al., 2018a; Mann and Davis, 2021).

Entomopathogenic fungi, particularly those belonging to the order Hypocreales (Ascomycota), act as natural insect antagonists in European forests (Wegensteiner et al., 2015a). Species of the genus *Beauveria* have repeatedly been identified as common pathogens within bark beetle populations, highlighting their efficacy as biocontrol agents (e.g. Landa et al., 2001; Kreutz et al., 2004a; Wegensteiner, 2007; Draganova et al., 2010; Takov et al., 2012; Mudrončeková et al., 2013; Wegensteiner et al., 2015a; Hyblerová et al., 2021). Among the five *Beauveria* species found in Europe (Rehner et al., 2011; Khonsanit et al., 2020; Kobmoo et al., 2021; Wang et al., 2022), four of them, *B. bassiana*, *B. caledonica*, *B. pseudobassiana* and *B. brongniartii*, have been documented in naturally encountered populations of bark beetles (e.g. Barta et al., 2018a, 2020). According to the findings of this study, infection by *B. bassiana* can be a natural mortality factor in *I. duplicatus*. *Beauveria bassiana* is generally the most prevalent fungal species in these populations and has been identified from various bark beetles. Most published studies in Europe have concentrated on investigating the effects of *B. bassiana* on *I. typographus* (e.g. Kreutz et al., 2004a, Grodzki and Kosibowicz, 2015, Barta et al., 2018b, 2020). Dinu et al. (2012) isolated a strain of *B. bassiana* from a naturally encountered population of *I. duplicatus* in Romania, demonstrating its potential against this pest, which has been recently confirmed under controlled laboratory settings (Vakula et al., 2019). One objective of this study was sampling and isolation of entomopathogenic fungi associated with *I. duplicatus* populations. Surprisingly, only one fungal species, *B. bassiana*, was isolated from the collected bark beetle specimens. This finding contrasts with previous research done on the closely related *I. typographus*, which has reported up so far four *Beauveria* species (e.g., Barta et al., 2018a, Hyblerová et al., 2021). Hypothetically, several factors could contribute to the observed discrepancy in fungal diversity between *I. duplicatus* and *I. typographus*. These may include environmental factors, host, fungal dynamics, and sampling methods. Factors in the environment such as

Table 8
Comparison of infection between *B. bassiana* strains and the time of carrier exposure after 12 days in Petri dishes based on the Tukey method for benchmarking family estimates. The log odds were exponentiated to obtain the odds ratios (OR).

Contrast	OR	2.50 %	97.50 %	SE	P-value	
NRID11/control	100	0.00	0.02	0.00	< 0.001	***
NRID43/control	2.63	0.17	0.87	0.13	< 0.02	*
NRID11/NRID43	51.39	23.39	112.91	17.26	< 0.001	***
A/B	2.78	1.23	6.29	0.88	0.007	**
A/C	28.57	10.55	77.39	11.08	< 0.001	***
A/D	114.32	33.58	389.17	54.51	< 0.001	***
B/C	10.26	4.17	25.25	3.60	< 0.001	***
B/D	41.07	13.04	129.33	18.34	< 0.001	***
C/D	4.00	1.49	10.77	1.54	0.002	**

habitat, temperature, humidity, and UV radiation can influence the distribution and prevalence of entomopathogenic fungi. It does not seem realistic that *B. bassiana* would be more tolerant of the environmental conditions in the studied *I. duplicatus* populations compared to other *Beauveria* species. Moreover, *I. duplicatus* shares the same habitat with *I. typographus*, thus these are the same for both species. *Beauveria pseudobassiana* exhibits a higher level of adaptation to forest ecosystems versus others, as evidenced by its documented preference for these habitats over agricultural or meadow environments (Medo et al., 2016). This fungus was recorded in *I. typographus* populations in Slovakia and was considered a common entomopathogen of bark beetles (Barta et al., 2018a, Hyblerová et al., 2021). In laboratory bioassays, this species showed high virulence against *Ips sexdentatus* (Boern.) and *I. typographus* demonstrating a capacity for being a promising biocontrol agent of these forest pests (Kocaçevik et al., 2016). Although *B. pseudobassiana* was not detected in the current survey, its potential presence in *I. duplicatus* can be anticipated as it shares the same ecological niche with *I. typographus*. *Ips duplicatus* might exhibit higher susceptibility to *B. bassiana* infection in comparison to other fungal species. However, further investigation is needed to explore the vulnerability of *I. duplicatus* to various *Beauveria* species and to conduct a more comprehensive sampling of the fungal communities associated with *I. duplicatus.* The potential bias in the sampling method employed in the study toward *B. bassiana* is debatable, as this procedure was used in prior studies involving *I. typographus* (Barta et al., 2018a, 2020; Hyblerová et al., 2021) where a greater diversity of fungi was identified.

Entomopathogenic fungi secrete hydrolytic exoenzymes like chitinase, protease, and lipase, which are used in the process of penetrating through the insect cuticle (Padmini and Padmaja, 2013; Ali and Moharram, 2014; Ferreira and Soares, 2023). These enzymes are essential for the infection mechanism and their production has been identified as a potential indicator of virulence for fungal isolates (Sánchez-Pérez et al., 2014; Mondal et al., 2016). Plate-based techniques are commonly employed to assess enzymatic activity in microorganisms, including fungi, and have also been adapted for use with entomopathogenic fungi (e.g. Bai et al., 2012; Padmini and Padmaja, 2013; Karimi et al., 2019). In the current research, the agar plate-based screening method of exocellular hydrolytic enzyme production was successfully used to identify highly virulent *B. bassiana* strains. Six strains with high enzymatic activity were selected for pathogenicity assessments, resulting in a significant impact on the survival of treated *I. duplicatus* adults. Except for one strain (NRID78), all strains exhibited a similar ($p < 0.05$) cumulative mortality rate in tested beetles 13 days post-treatment compared to the highly virulent strain DSM 32081 (a reference strain) previously selected for use against *I. typographus*.

Conifer secondary metabolites, such as monoterpenes, play a crucial role in the tree's defense system and can inhibit fungal pathogenicity. This suggests that certain monoterpenes may eliminate specific strains of *B. bassiana*, thereby preventing the development of bark beetle infection following colonization of conifer phloem (Mann and Davis, 2021). For our experiments, we selected five monoterpenes found in the phloem of Norway spruce *P. abies* (Marešová et al., 2022). All tested monoterpenes exhibited inhibitory effects on the radial growth of *B. bassiana* strains *in vitro*, consistent with a recent study by Mann and Davis (2020). While limonene completely inhibited the growth of the strains, the effect of pinene was less pronounced and varied among the strains. Notably, the NRID11 and NRID43 strains responded positively to pinene. Research conducted by Mann and Davis (2020) demonstrated that monoterpenes derived from Engelmann spruce have strong inhibitory effects on the growth of *B. bassiana*. α-Pinene had a moderate inhibitory impact, reducing average colony growth by 71 %, while myrcene and β-pinene were the least effective, reducing the mean radius of colony by 58 % and 57 %, respectively. Although the *in vitro* effects of limonene on *B. bassiana* growth are evident in our results, the *in vivo* dynamics within bark beetle galleries may be more complex, and thus, the activity of limonene in galleries may not fully reflect the results seen *in vitro*. This is supported by the observation that *Beauveria* infections are normally found in bark beetles within the galleries of Norway spruce trees. Several hypothetical factors could potentially influence the inhibitory effects of limonene under natural conditions. Micro-environmental variability (1): limonene concentrations may vary within different parts of the bark beetle gallery, potentially creating micro-niches where fungal growth is less inhibited. Fungal adaptation (2): some *B. bassiana* strains may have developed mechanisms to tolerate the presence of limonene or other monoterpenes to some extent. A recent study demonstrated variable tolerance to pine monoterpenes in *B. bassiana* strains (Rosana et al., 2021). The study identified three phenotypic groups of *B. bassiana* with different characteristics, including their ability to withstand various monoterpenes such as limonene. Mycobiome complexity (3): other microorganisms present within bark beetle galleries might influence the effect of limonene on *B. bassiana*). For example, *Grosmannia clavigera* (Rob.-Jeffr. & R.W. Davidson) Zipfel, Z.W. de Beer & M.J. Wingf., the main fungal pathogen associated with the mountain pine beetle (*Dendroctonus ponderosae* Hopkins), possesses specialized enzymes that can modify and degrade host tree monoterpenes, particularly (+)-limonene (Wang et al., 2014). It is important to note that further controlled studies are needed to empirically test these hypotheses and elucidate the actual interactions between *B. bassiana*, limonene (or other monoterpenes), and the bark beetle gallery environment.

In the field experiments, we employed modified pheromone traps with inoculated carriers, also referred to as an "assisted autodissemination device", to introduce selected strains of *B. bassiana* to adult *I. duplicatus*. We used a trap applicator of fungal inoculum with a patented carrier containing *B. bassiana*. When comparing the results after the 6-day and 12-day periods, the overall patterns remained similar, with strain NRID11 exhibiting the strongest effect on infection, and the upfront phase A (3 days in traps) of carrier exposure having the highest impact. Strain NRID43 did not show significant differences compared to the control. While we were working with increased caution when manipulating the imagines, we were not able to prevent the infection of control imagines. Similar experiences were reported by Kreutz et al. (2004a), indicating the highly contagious nature of the pathogen. This was most notable in the initial period of the experiment when the fungal strains were most active. The duration of carrier exposure in the trap influenced the likelihood of infection. Prevalence decreased as the time of carrier exposure increased from phase A (3 days in traps) to phase D (60 days in traps). This implies that the *Beauveria* inoculum in the carrier lost its viability over time, as the exposure period prolonged. This outcome was anticipated and has been corroborated by other studies. For instance, in previous research, carriers of *B. bassiana* remained active for 42 days with an efficiency of 80 %; however, by 56 days, it decreased to 55 % (Rell et al., 2022). There was no significant difference in the likelihood of infection between phases C (21 days) and D (30 days), suggesting that the rate may plateau at a certain level of exposure. This plateau might indicate that after some time, the remaining viable conidia are insufficient to increase infection rates significantly. Based on the results, the carrier was able to preserve virulence for 21 days. These observations are from a single-season experiment in 2022, during which the weather was very hot and dry, negatively affecting the carrier's ability to maintain the fungus active within it. High temperatures and low humidity are known to reduce the viability of fungal conidia (Cabanillas and Jones, 2009). Therefore, environmental conditions likely played a role in the reduced efficacy over time. Lalík et al. (2021) noted that several factors affect the survival of fungi in the carrier. The carriers with entomopathogenic fungi were exposed to UVC radiation for 120 h and freezing at −18 °C. These carrier materials stressed in this way were able to infect and kill *Hylobius abietis* (L.) adults just as well as fresh ones. However, they did not investigate longer exposure to stress. This suggests that while the carrier can protect the fungi from certain negative factors for a limited time, prolonged exposure, especially under adverse environmental conditions, can diminish its effectiveness. Overall, our findings highlight how important is to optimize the time of carrier exposure and to select highly virulent fungal strains that are environmentally resilient for utilization this approach in the biological control of *I. duplicatus*.

Kreutz et al. (2004a) demonstrated that the transmission of *B. bassiana* between bark beetles, as well as the dissemination of the fungus within a bark beetle population, is possible through modifications to the collecting boxes in pheromone traps. The efficient activation of *Beauveria* spores on individuals under bark may be influenced by environmental conditions or the susceptibility of beetles to infection (Vakula et al., 2019), which in turn is affected by their health or exposure to stress factors. However, more research is needed to understand the mechanisms of horizontal and vertical transmission of the infection between individuals present in galleries found under the bark.

5. Conclusions

We isolated local *B. bassiana* strains from *I. duplicatus* and assessed their potential for biological control. Our findings indicate that *B. bassiana* is naturally associated with this beetle species, and the fungal strains exhibited varying enzymatic activities and sensitivities to spruce monoterpenes *in vitro*. Pathogenicity tests identified strain NRID11 as highly virulent against adult *I. duplicatus* under laboratory conditions. Field experiments demonstrated the potential of modified pheromone traps, equipped with a patented carrier inoculated with *B. bassiana* (strain NRID11), to deliver fungal inoculum to bark beetles. However, the results also showed that the efficacy of the inoculum in the carrier decreased over time.

While our findings indicate that *I. duplicatus* can be successfully inoculated with *B. bassiana* using this delivery system, we want to emphasize the need for further research to fully understand the complex interactions among *B. bassiana*, host trees, and the bark beetle microbiome in natural environments. Subsequent field studies should focus on the timing of inoculum delivery, selection of fungal strains, and formulation to enhance field persistence and virulence. Additionally, multi-season trials and analyses of fungal dissemination are essential to thoroughly evaluate the feasibility and efficacy of this approach. In conclusion, this study provides a foundation for further exploration of *B. bassiana* as a potential component of integrated pest management strategies for *I. duplicatus*.

CRediT authorship contribution statement

Jozef Vakula: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Christo Nikolov:** Visualization, Data curation. **Michal Lalík:** Methodology, Data curation. **Miriam Kádasi Horáková:** Investigation, Data curation. **Slavomír Rell:** Data curation. **Juraj Galko:** Writing – review & editing, Data curation. **Andrej Gubka:** Formal analysis, Data curation. **Milan Zúbrik:** Writing – review & editing, Formal analysis. **Andrej Kunca:** Formal analysis, Data curation. **Marek Barta:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Funding acquisition, Data curation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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